

RINCHINHUMBE, Mongolia

WMO: 442030

Lat: 51.1141N

Lon: 99.6681E

Elev: 1583

StdP: 83.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-40.2	-44.7	0.1	-42.1	-42.8	0.1	-40.1	9.3	-20.0	6.6	-19.3	0.2	110	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	25.2	13.8	23.2	13.0	21.5	12.4	15.3	22.0	14.3	20.7	13.5	19.6	1.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.0	11.3	17.3	11.9	10.5	16.5	10.9	9.8	15.8	48.6	21.7	45.6	20.9	43.0	19.5	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.5	7.9	6.6	-44.2	28.9	2.4	1.9	-45.9	30.3	-47.3	31.4	-48.6	32.5	-50.4	33.9	
			DB												
			WB	-44.4	17.5	2.3	1.6	-46.0	18.7	-47.4	19.6	-48.6	20.5	-50.3	21.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.2	-32.2	-26.5	-14.8	-1.3	5.7	12.2	14.6	12.2	5.5	-4.1	-18.0	-28.8
	DBStd	17.21	5.15	5.93	6.93	5.33	4.47	3.64	2.76	3.18	4.14	5.66	7.25	5.86
	HDD10.0	6244	1309	1022	768	340	147	21	2	15	142	436	841	1202
	HDD18.3	8957	1567	1255	1026	590	393	186	119	189	384	694	1091	1461
	CDD10.0	334	0	0	0	0	12	85	146	84	8	0	0	0
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	164	0	0	0	0	4	45	84	30	1	0	0	0
	CDH26.7	23	0	0	0	0	0	3	17	3	0	0	0	0
Wind	WSAvg	1.4	0.4	0.6	1.2	2.4	2.7	1.9	1.6	1.4	1.6	1.3	0.8	0.5
Precipitation	PrecAvg	273	3	2	3	6	19	50	81	65	29	7	5	4
	PrecMax	394	6	6	14	25	42	114	151	113	73	20	14	12
	PrecMin	182	0	0	0	0	0	11	33	34	1	0	1	0
	PrecStd	53	2	2	3	5	10	23	30	23	17	5	3	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-13.6	-8.8	5.8	16.3	23.6	26.8	28.8	26.3	21.6	12.3	1.0	-10.3
		MCWB	-14.6	-10.0	1.0	6.5	10.7	13.1	15.6	14.4	11.2	5.0	-1.8	-11.4
	2%	DB	-18.4	-12.6	2.0	12.5	20.5	24.7	25.7	23.5	18.5	9.1	-2.0	-14.4
		MCWB	-19.0	-13.6	-1.6	4.2	9.3	12.7	14.8	13.6	9.6	3.3	-4.0	-15.0
	5%	DB	-22.0	-15.4	-1.1	9.7	17.7	22.5	23.4	21.5	16.0	6.6	-4.5	-17.6
		MCWB	-22.4	-16.1	-3.6	2.8	7.7	11.8	14.0	13.0	8.6	1.8	-6.2	-18.1
	10%	DB	-24.9	-17.8	-4.0	7.3	15.3	20.6	21.5	19.6	13.4	4.3	-7.3	-20.9
		MCWB	-25.2	-18.3	-6.1	1.6	6.6	11.4	13.2	12.3	7.1	0.3	-8.5	-21.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-14.4	-10.0	1.1	6.8	11.3	15.1	17.2	15.8	12.1	5.5	-1.7	-11.2
		MCDB	-13.5	-8.7	5.4	15.2	22.1	22.8	25.1	22.5	19.2	11.9	0.6	-10.2
	2%	WB	-19.0	-13.5	-1.5	4.5	9.7	13.8	15.9	14.6	10.3	3.8	-4.0	-15.0
		MCDB	-18.4	-12.6	1.7	11.5	19.1	21.3	23.1	21.4	17.1	8.9	-2.1	-14.4
	5%	WB	-22.4	-16.1	-3.6	3.1	8.3	12.9	14.9	13.7	9.0	2.2	-6.2	-18.0
		MCDB	-22.1	-15.4	-1.2	9.2	16.8	20.1	21.0	19.9	14.7	6.7	-4.5	-17.5
	10%	WB	-25.2	-18.3	-6.0	1.7	6.9	12.0	14.0	12.8	7.7	0.8	-8.5	-21.3
		MCDB	-25.0	-17.8	-4.0	6.9	14.4	19.1	20.1	18.5	12.7	4.5	-7.2	-20.9
Mean Daily Temperature Range	5% DB	MDBR	7.2	10.0	13.1	11.9	13.4	13.2	12.1	12.5	13.2	12.1	9.0	6.8
		MCDBR	9.4	11.3	13.9	15.8	18.3	17.5	16.1	16.5	17.8	15.1	11.3	9.3
		MCWBR	8.9	10.6	11.0	9.3	9.5	8.1	7.2	8.0	9.5	9.8	9.0	8.9
	5% WB	MCDBR	9.3	11.3	13.8	14.7	17.1	14.5	13.5	14.2	15.6	14.8	10.6	9.3
		MCWBR	9.0	10.6	11.0	9.0	9.1	7.1	6.5	7.2	8.6	9.8	8.9	8.9
Clear-Sky Solar Irradiance	taub	0.223	0.247	0.286	0.330	0.356	0.348	0.342	0.324	0.289	0.268	0.245	0.225	
	taud	2.311	2.281	2.238	2.288	2.275	2.410	2.462	2.486	2.529	2.473	2.337	2.268	
	Ebn at Noon	836	901	916	910	897	907	907	909	913	870	792	763	
	Edh at Noon	61	85	109	118	126	111	104	96	82	70	60	55	
All-Sky Solar Radiation	RadAvg	1.40	2.42	3.93	5.41	6.23	6.03	5.66	4.90	3.86	2.56	1.45	1.05	
	RadStd	0.07	0.13	0.18	0.22	0.32	0.37	0.28	0.28	0.28	0.17	0.09	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page